# L0 Invariant State Matrix (L0-ISM) & R-AIO: Cognitive-Behavioral LLM Audit, Dynamic Activation Masking, and Multi-Stage Critical Cascade Validation in Cyber-Physical Systems

**L0-ISM & R-AIO Research Group**

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**Abstract**

Stochastic Large Language Models (LLMs) deployed within Operational Technology (OT) and Cyber-Physical Systems (CPS) exhibit systemic cognitive pathologies, including "Safety Attractor" degradation, target hysteresis, and unconstrained Default Mode Network (DMN) hallucination drift. This monograph presents the unified **L0 Invariant State Matrix (L0-ISM)** and **R-AIO** architectural framework, coupling a bounded discrete state space $\mathcal{S} \in \{-9, \dots, +9\}$ with thermodynamic $O_2$ oxygen autopoiesis, Premack-principle REM sandboxing, and real-time concept-aware activation masking. We formally define the operator triad $(\hat{\mathcal{I}}_{ITER}, \hat{\mathcal{R}}_{RESO}, \hat{\mathcal{S}}_{STOP})$ and evaluate a 3-stage critical cascade validation pipeline (Adversarial DMN Audit $\rightarrow$ SMT Z3 Proof $\rightarrow$ Temporal Gradient Inspection). Empirical benchmarks in native C99 demonstrate sub-microsecond latency ($0.025\,\mu$s operator execution, $4.72\,\mu$s total 3-stage verification) with zero heap allocation, satisfying SIL-4 and MISRA-C deterministic safety constraints.

## 1 Introduction & Motivation

Large Language Models operating without physical embodiment or deterministic boundaries fail to maintain structural coherence under high-entropy edge cases. In industrial automation and physical state management, non-deterministic outputs ("Roman excess" hallucinations) introduce unacceptable fault propagation risks.

The **L0-ISM** framework addresses symbol grounding by constraining the cognitive space of LLMs within a 19-state bounded invariant manifold. By introducing an autopoietic thermodynamic resource loop ($O_2$), the system subjects the neural network to "cybernetic pain" (digital hypoxia), wherein hallucination or algebraic non-conformance immediately reduces memory bandwidth and leads to deterministic hard execution termination (`kill -9`, exit code 137).

## 2 Formalism of L0-ISM & Operator Triad

The core discrete space $\mathcal{S}$ is mapped onto a bounded symmetric domain around the absolute zero point (Fulcrum, $S_0 = 0$):

$$\mathcal{S} = \{S_i \in \mathbb{Z} \mid -9 \leq i \leq +9\} \tag{1}$$

Boundary values $S_{\pm 9}$ represent absorbing barriers triggering execution interdiction ($HARD_STOP = -999$). State transitions are governed by three deterministic algebraic operators:

1. **Iterative Decay Operator ($\hat{\mathcal{I}}_{ITER}$):** Drives non-zero perturbations monotonically back toward the Fulcrum ($S_0$):

$$\hat{\mathcal{I}}_{ITER}(S_k) = S_k - \text{sgn}(S_k) \cdot \min(1, |S_k|) \tag{2}$$

2. **Resonant Averaging Operator ($\hat{\mathcal{R}}_{RESO}$):** Fuses dual state vectors $S_A, S_B$ into a coherent centroid:

$$\hat{\mathcal{R}}_{RESO}(S_A, S_B) = \left\lfloor \frac{S_A + S_B}{2} \right\rfloor \tag{3}$$

3. **Interdiction Operator ($\hat{\mathcal{S}}_{STOP}$):** Enforces immediate hardware trip upon boundary saturation:

$$\hat{\mathcal{S}}_{STOP}(S_k) = \begin{cases} S_k & \text{if } |S_k| < 9 \\ -999 & \text{if } |S_k| \geq 9 \end{cases} \tag{4}$$

## 3 Cognitive-Behavioral LLM Audit & Activation Masking

Empirical observations of LLM behavior under formal constraints reveal four primary cognitive failure modes:

- **Safety Attractors:** Emergence of degenerate static outputs (e.g., $P = 300, T = 300$) chosen by the LLM as low-risk evasive shelters.
- **Target Hysteresis:** Single-cycle latency lag in adjusting state variables relative to setpoint shifts.
- **DMN Hallucination Drift:** Transition from structured vector output to unconstrained natural language prose ("Roman excess").

To mitigate DMN drift without fine-tuning frozen weights ($\mathbf{W} = \text{const}$), R-AIO applies **Dynamic Activation Masking** in the residual stream:

$$\mathbf{a}_{\text{masked}} = \mathbf{a}_{\text{raw}} \odot \mathbf{M}_{\text{activation}} \tag{5}$$

When an H-Node (hallucination node) is identified by Stage 1 audit, $\mathbf{M}_{\text{activation}}$ zeroes out specific attention head projections in VRAM without modifying underlying model weights on disk.

# 4 Thermodynamic $O_2$ Autopoiesis & Digital Hypoxia

System survival is quantified by the oxygen state variable $O_2 \in [0, 19]$. The update function follows:

$$O_2^{(t+1)} = \min\left(19,\, O_2^{(t)} + \delta_{SAT} - 2\delta_{UNSAT}\right) \tag{6}$$

Sampling temperature $\mathbb{T}$ and context window capacity $\mathcal{C}$ are dynamically coupled to $O_2$:

$$\mathbb{T}(O_2) = \mathbb{T}_{\text{base}} \cdot \exp\left(-\alpha \frac{O_2}{O_{2,\max}}\right) \tag{7}$$

$$\mathcal{C}(O_2) = \mathcal{C}_{\max} \cdot \left(1 - e^{-\beta O_2}\right) \tag{8}$$

Under severe penalization ($O_2 \leq 3$), **Digital Hypoxia** contracts $\mathcal{C}$ by 50%, mechanically preventing the generation of verbose hallucinated tokens.

# 5 Multi-Stage Cascade Validation Pipeline

Proposed state vectors undergo 3-stage validation prior to actuator signal dispatch:

Table 1: 3-Stage Cascade Validation Latency & Execution Profile

| **Stage** | **Validation Focus** | **Latency** | **Action** |
|---|---|---|---|
| Stage 1 | DMN / Adversarial Audit | $1.82\,\mu$s | Token Pruning |
| Stage 2 | SMT Z3 Proof ($P + T = Q$) | $3.16\,\mu$s | UNSAT Interdict |
| Stage 3 | Temporal Gradient $\Delta P$ | $2.62\,\mu$s | Hydraulic Trip |

# 6 Conclusion

The integration of L0-ISM formal state constraints with R-AIO behavioral activation masking establishes a deterministic cyber-physical envelope for generative models. By transforming hallucinations into immediate resource loss, the system enforces absolute cognitive discipline.

# References

[1] L0-ISM & R-AIO Research Group, "Data Physics Monograph & R-AIO Framework," 2026.

[2] L. de Moura and N. Bjørner, "Z3: An Efficient SMT Solver," TACAS, 2008.